

8.2.24

EE25BTECH11057 - Rushil Shanmukha Srinivas

Question : The conic has vertices $(\pm 6, 0)$ and foci $(\pm 4, 0)$. Find the equation of the conic.

Solution :

Name	Description	vector form
$\mathbf{B}_1$	vertex 1 of conic	$\begin{pmatrix} 6 \\ 0 \end{pmatrix}$
$\mathbf{B}_2$	vertex 2 of conic	$\begin{pmatrix} -6 \\ 0 \end{pmatrix}$
$\mathbf{F}_1$	focus 1 of conic	$\begin{pmatrix} 4 \\ 0 \end{pmatrix}$
$\mathbf{F}_2$	focus 2 of conic	$\begin{pmatrix} -4 \\ 0 \end{pmatrix}$

Table : Ellipse

The conic has two foci , so it cannot be a parabola .

Equation for any conic with directrix $\mathbf{n}^\top \mathbf{x} = c$, eccentricity e and focus $\mathbf{F}$ is given by

$$\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \quad (0.1)$$

$$\mathbf{V} = \|\mathbf{n}\|^2 \mathbf{I} - e^2 \mathbf{n} \mathbf{n}^\top \quad (0.2)$$

$$\mathbf{u} = ce^2 \mathbf{n} - \|\mathbf{n}\|^2 \mathbf{F} \quad (0.3)$$

$$f = \|\mathbf{n}\|^2 \|\mathbf{F}\|^2 - c^2 e^2 \quad (0.4)$$

The normal vector of the directrix is along the direction vector of $\mathbf{F}_1 - \mathbf{F}_2$

$$\mathbf{n} = \mathbf{F}_1 - \mathbf{F}_2 \equiv \mathbf{e}_1 \quad (0.5)$$

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (0.6)$$

From (0.2) we can form the matrix $\mathbf{V}$

$$\mathbf{V} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - e^2 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad (0.7)$$

$$\mathbf{V} = \begin{pmatrix} 1 - e^2 & 0 \\ 0 & 1 \end{pmatrix} \quad (0.8)$$

As $\mathbf{V}$ is an upper triangular matrix, we get the eigen values as the diagonal entries

$$\lambda_1 = 1 - e^2 \quad \lambda_2 = 1 \quad (0.9)$$

Clearly $|\mathbf{V}| \neq 0$, $\mathbf{V}^{-1}$ exists.

The center of the conic $\mathbf{c}$ can be found

$$\mathbf{c} = \frac{\mathbf{F}_1 + \mathbf{F}_2}{2} = \mathbf{0} \quad (0.10)$$

The relation between the $\mathbf{c}$, $\mathbf{V}$ and $\mathbf{u}$ is given by

$$\mathbf{V}\mathbf{c} + \mathbf{u} = \mathbf{0} \quad |\mathbf{V}| \neq 0 \quad (0.11)$$

$$\mathbf{c} = \mathbf{0} \quad (0.12)$$

$$\mathbf{u} = \mathbf{0} \quad (0.13)$$

From (0.3) we get

$$ce^2 \mathbf{e}_1 = \mathbf{F}_1 \quad (0.14)$$

$$\begin{pmatrix} ce^2 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} \quad (0.15)$$

$$ce^2 = 4 \quad (0.16)$$

$$c = \frac{4}{e^2} \quad (0.17)$$

$$f_0 = \mathbf{u}^\top \mathbf{V}^{-1} \mathbf{u} - f \quad (0.18)$$

as $\mathbf{u} = \mathbf{0}$ and from (0.4), we get

$$f_0 = c^2 e^2 - 16 \quad (0.19)$$

The length of the major axis is the distance between the two vertices

$$\|\mathbf{B}_1 - \mathbf{B}_2\| = 12 \quad (0.20)$$

The length of major axes is also given as

$$2\sqrt{\left|\frac{f_0}{\lambda_1}\right|} \quad (0.21)$$

So,

$$2\sqrt{\left|\frac{c^2e^2 - 16}{1 - e^2}\right|} = 12 \quad (0.22)$$

From (0.17) we get

$$\sqrt{\frac{16}{e^2}} = 6 \quad (0.23)$$

$$\frac{4}{e} = 6 \quad (0.24)$$

$$e = \frac{2}{3} \quad (0.25)$$

As $e < 1$, the conic is an **ellipse**

The value of c and directrix equation are given as

$$c = 9 \quad \mathbf{n}^\top \mathbf{x} = \pm 9 \quad (0.26)$$

Using the obtained values of c and e , we get

$$\mathbf{V} = \begin{pmatrix} \frac{5}{9} & 0 \\ 0 & 1 \end{pmatrix} \quad (0.27)$$

$$\mathbf{u} = \mathbf{0} \quad (0.28)$$

$$f = -20 \quad (0.29)$$

Substituting these in (0.1) , we get the equation of **ellipse** as

$$\mathbf{x}^\top \begin{pmatrix} \frac{1}{36} & 0 \\ 0 & \frac{1}{20} \end{pmatrix} \mathbf{x} = 1 \quad (0.30)$$

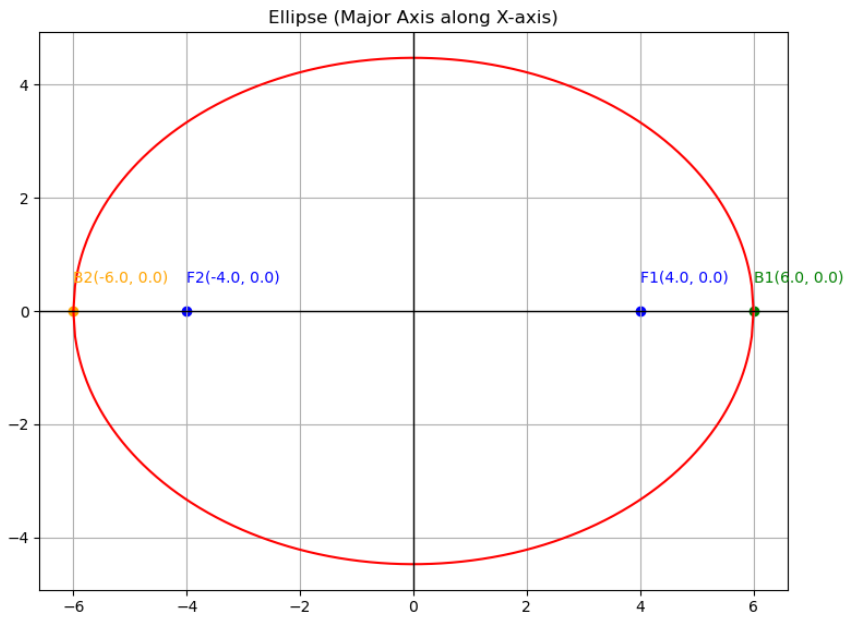


Fig : Ellipse